

Lecture 3: Introduction to R and RStudio

Part 1: `tidyverse` (follow along in RStudio) Academic Year
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Learning Outcomes

What you will learn in this session:

- The Structure of R commands
- About the `tidyverse` package for data frames
 - `select()` and rename columns (variables)
 - `filter()` rows (observations)
 - `mutate()` (define columns (variables); overwrite old or create new)
 - piping (connecting commands) with `|>`.

Basics of R Commands

R commands consists of the **command's name followed by a pair of parentheses**:

```
command()
```

Inside the () we can define one or more **arguments** for the command.

```
VecTest=c(1,2,3)
```

```
sum(x=VecTest)
```

```
[1] 6
```

```
mean(VecTest)
```

```
[1] 2
```

- Arguments in a command usually have names such as `x=` or `data=`
- R does not require to use the argument's name, but **order matters**
- R commands have many arguments. Most have default values
- We can nest commands. However, nesting too deeply makes code difficult to read.

Structure of R Commands

Most R commands have the following structure:

Often the `data` argument is the first argument in a command.

Usually named `data=` or `x=`.

Use a Command with and without Argument Names

```
VecTest=c(1,2,3)
Result=mean(x=VecTest, trim=0, na.rm=FALSE)
cat("The mean of the values in vector VecTest is:", Result)
The mean of the values in vector VecTest is: 2
```

```
...
Result=mean(VecTest, 0, FALSE)
cat("The mean of the values in vector VecTest is:", Result)
The mean of the values in vector VecTest is: 2
```

```
...
Result=mean(VecTest)
cat("The mean of the values in vector VecTest is:", Result)
The mean of the values in vector VecTest is: 2
```

...

All three examples are equivalent

Try ? mean in the Rstudio console to see the default values.

Important Commands from `tidyverse/dplyr` Package

- `dplyr` package is part of the `tidyverse` (meta) package
- `library(tidyverse)` (loads the `tidyverse` and its packages)
- `select()` selects columns (variables) from a data frame
- `filter()` filters rows (observations) for specific criteria
- `mutate()` calculates new or overwrites existing columns (variables) based on other columns (just like Excel).

Titanic Dataset

```
library(rio)
DataTitanic=import("Data/Titanic.csv")
head(DataTitanic)
```

Survived	Passenger Class	Name	Gender	Age	Siblings/Spouses Aboard	Parents/Children Aboard	Fare in Pounds
0	3	Mr. Owen Harris Braund	male	22	1	0	7.2500
1	1	Mrs. John Bradley Cumings	female	38	1	0	71.2833
1	3	Miss. Laina Heikkinen	female	26	0	0	7.9250
1	1	Mrs. Jacques Heath Futrelle	female	35	1	0	53.1000
0	3	Mr. William Henry Allen	male	35	0	0	8.0500
0	3	Mr. James Moran	male	27	0	0	8.4583

The `select()` Command

`select(DataMine, Var1, Var2)` selects columns (variables) `Var1` and `Var2` from a data frame `DataMine`. The first argument is the `data=` argument followed by the names of the selected variables.

`select(Data, -Var1, -Var2)` selects all columns (variables) except `Var1` and `Var2` from a data frame `DataMine`.

...

Here is an example using the `DataTitanic` data frame from the previous slide:

```
library(tidyverse)
DataTitanicSelVar=select(DataTitanic, Survived, Name, Gender, Age)
head(DataTitanicSelVar)
```

Survived	Name	Gender	Age
0	Mr. Owen Harris Braund	male	22
1	Mrs. John Bradley Cumings	female	38
1	Miss. Laina Heikkinen	female	26
1	Mrs. Jacques Heath Futrelle	female	35
0	Mr. William Henry Allen	male	35
0	Mr. James Moran	male	27

The `filter()` Command

The `filter()` command filters rows (observations) of a data frame for specific criteria. The first argument is the `data=` argument followed by the filter criteria.

...

E.g., *filter* for female passengers from the dataset: Use `DataTitanicSelVar` that we created in the previous slide (note that we have to use `==` instead of `=` for the criteria):

```
DataTitanicSelVarFem=filter(DataTitanicSelVar, Gender=="female")  
head(DataTitanicSelVarFem)
```

Survived	Name	Gender	Age
1	Mrs. John Bradley Cumings	female	38
1	Miss. Laina Heikkinen	female	26
1	Mrs. Jacques Heath Futrelle	female	35
1	Mrs. Oscar W Johnson	female	27
1	Mrs. Nicholas (Adele Achem) Nasser	female	14
1	Miss. Marguerite Rut Sandstrom	female	4

The mutate () Command

`mutate()` creates or overwrites columns (variables) based on other columns (just like Excel). The first argument is the `data=` argument followed by the instructions on how to create the new variable.

...

E.g., *mutate* calculates new column `Born` based on `Age` during Titanic disaster (1912). Uses `DataTitanicSelVarFem` from previous slide:
`DataTitanicSelVarFemBirthYear=mutate(DataTitanicSelVarFem, Born=1912-Age)`
`head(DataTitanicSelVarFemBirthYear)`

Survived	Name	Gender	Age	Born
1	Mrs. John Bradley Cumings	female	38	1874
1	Miss. Laina Heikkinen	female	26	1886
1	Mrs. Jacques Heath Futrelle	female	35	1877
1	Mrs. Oscar W Johnson	female	27	1885
1	Mrs. Nicholas (Adele Achem) Nasser	female	14	1898
1	Miss. Marguerite Rut Sandstrom	female	4	1908

Summary

1. We selected variables *Survived, Name, Gender, Age* and saved in `DataTitanicSelVar`
2. We filtered for females and saved in `DataTitanicSelVarFem`
3. We mutated to calculate new variable and saved finally in `DataTitanicSelVarFemBirthYear`

Could this be done easier?

Note, overwriting data frames such as `DataTitanic` is usually a bad idea!

Alternative: Nesting

(I am not serious)

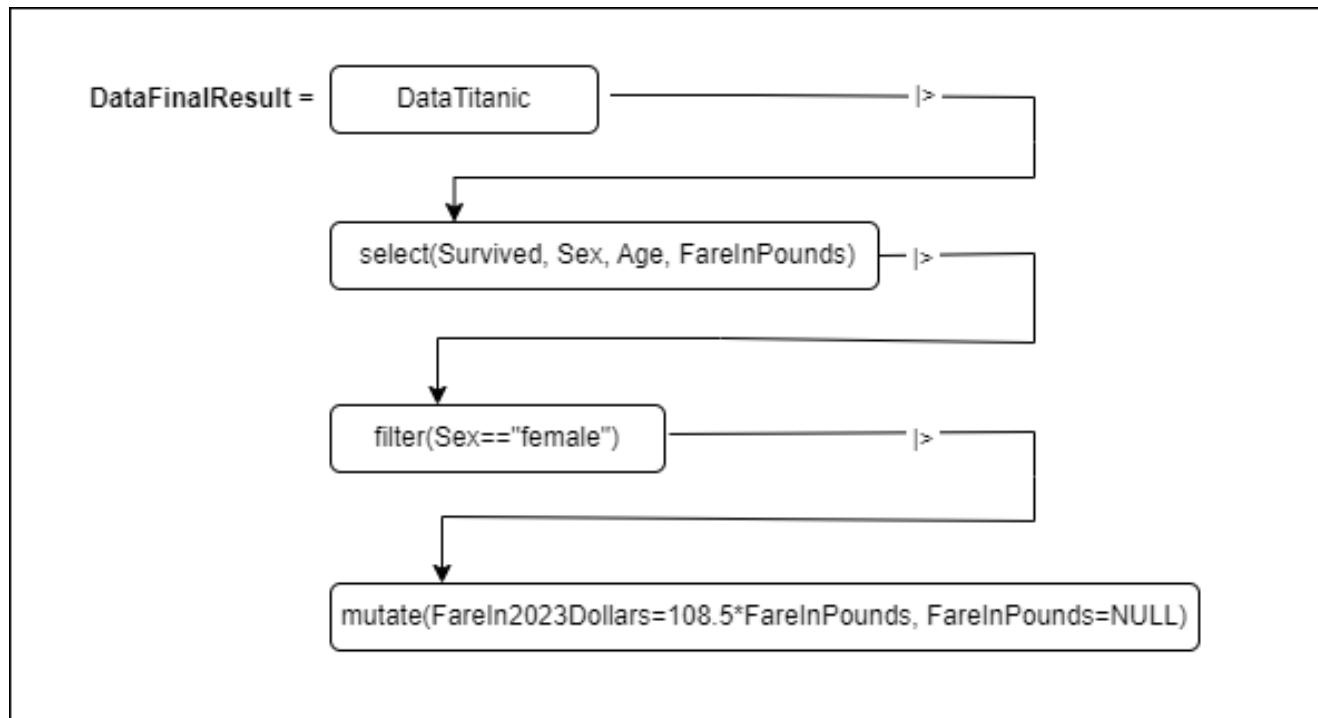
```
library(tidyverse)
DataTitanicFinal= mutate(

filter(select(DataTitanic, Survived,
Name, Gender, Age),
      Gender=="female"),
      Born=1912-
Age)
head(DataTitanicFinal)
```

Survived	Name	Gender	Age	Born
1	Mrs. John Bradley Cumings	female	38	1874
1	Miss. Laina Heikkinen	female	26	1886
1	Mrs. Jacques Heath Futrelle	female	35	1877
1	Mrs. Oscar W Johnson	female	27	1885
1	Mrs. Nicholas (Adele Achem) Nasser	female	14	1898
1	Miss. Marguerite Rut Sandstrom	female	4	1908

Piping Schema

Piping Schema



Alternative: Piping

(will be used throughout the course/book)

```
library(tidyverse)
DataTitanicFinal = DataTitanic |>
  select(Survived,
         Name, Gender, Age) |>
  filter(Gender=="female") |>
  mutate(Born=1912-
         Age)
head(DataTitanicFinal)
```

Survived	Name	Gender	Age	Born
1	Mrs. John Bradley Cumings	female	38	1874
1	Miss. Laina Heikkinen	female	26	1886
1	Mrs. Jacques Heath Futrelle	female	35	1877
1	Mrs. Oscar W Johnson	female	27	1885
1	Mrs. Nicholas (Adele Achem) Nasser	female	14	1898
1	Miss. Marguerite Rut Sandstrom	female	4	1908

Questions